



A comprehensive parametric study on NO and N₂O formation in ammonia-methane cofired premixed flames: Spatially resolved measurements and kinetic analysis

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ABSTRACT

Understanding the mechanism of NO_x formation and destruction is a prerequisite for the development of effective NO_x mitigation techniques in ammonia flames. Laboratory-scale laminar ammonia (NH₃) flames are well suited for such fundamental kinetic studies as the complex interaction between chemistry and fluid flow can be largely decoupled. However, quantitative and spatially resolved NO/N₂O concentration data in canonical laminar ammonia flames are surprisingly scarce. Such data is, on the other hand, crucial for developing and validating kinetic models for NH₃ combustion. In this regard, we developed a novel NO/N₂O measurement method combining microprobe sampling and calibration-free mid-infrared laser absorption spectroscopy and realized spatially-resolved detection with high accuracy and large dynamic ratio. The fidelity of the method has been rigorously tested before being applied to perform a comprehensive parametric study on NO and N₂O formation in NH₃-CH₄ co-fired burner-stabilized premixed flames. The effects of NH₃ fuel ratio, equivalence ratio and flame temperature on NO/N₂O formation have been experimentally determined. Corresponding numerical modelling was also performed using literature-based mechanisms to provide kinetic insights into the experimental observations. It is found that existing mechanisms generally have satisfactory predictions in fuel-lean flames; however, under fuel-rich conditions, these mechanisms overpredict NO formation but underestimate its destruction, leading to significant over-prediction. Such discrepancies were further investigated through sensitivity and reaction pathway analysis. The present study not only provides extensive spatially-resolved NO/N₂O data in ammonia flames that are urgently needed for the development and validation of NH₃ combustion mechanisms, the comparison between neat CH₄ and NH₃-cofired flames also points to the fact that traditional NO_x mitigation techniques that are popular in combustion of hydrocarbon fuels may not be appropriate in NH₃ flames. Perhaps most interestingly, the present experimental results show an oxygen-enriched oxidizer can in some cases reduce NO emission from NH₃ flames.

1. Introduction

Combustion applications of NH₃ have recently attracted significant research attention [1–6]. The results of these studies have revealed two major concerns regarding NH₃ combustion [5] despite its benefits of being CO₂ emission-free. First, due to the low reactivity of the molecule, NH₃ flames can be subject to ignition difficulties, flame instability issues as well as significant fuel slip. Second, fuel-bound nitrogen in the NH₃ molecule can result in excessive NO_x emissions. A variety of techniques

have been developed to address these two concerns. For the former, fuel blending with more reactive fuels such as hydrogen [7], natural gas [8–10], and certain oxygenated fuels [11], partial NH₃ cracking [12], oxygen enriched combustion [13,14], swirling [15], plasma-assisted [16] and super-adiabatic combustion [17] have been proven to be among the effective approaches. For the NO_x emission issues, methods of staged combustion [18,19], MILD combustion [20,21], plasma treatments [22] and humidification [23] have been tested. It can be generally remarked that challenges on flame stability have been better

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tackled, as evidenced by a number of studies reporting successful stabilization of laboratory-scale [17] and even industrial-scale [3] neat NH_3 /air flames. On the other hand, a sampling of the literature data reveals that NH_3 combustion can have NO emissions 1 to 2 orders of magnitude higher than those of conventional hydrocarbon flames so that it is extremely challenging to keep exhaust NO concentrations below regulatory values [24]. Kurata et al. [18] reported that even with an optimized rich-lean staged combustion strategy, the minimum NO emissions from an NH_3 -fired micro gas turbine can still be as high as 337 ppm (16 % O_2)—nearly 5 times higher than the Japanese regulation at 70 ppm. Clearly, future innovations on NOx emission mitigation are needed.

An understanding of NH_3 oxidation and NO formation kinetics is a prerequisite for developing low-NOx NH_3 combustion technologies. The community has devoted significant efforts in this direction. Global parameters such as ignition delay time, extinction strain rate and laminar flame speed have been respectively measured in shock tube [25] / rapid compression machines [26], spherical expanding flame [13] / heat flux burner [27], and counterflow flames [28], to assist the development of detailed NH_3 reaction mechanisms. Quantitative mole fractions for important species including NO have also been measured in idealized reactors such as shock tube [29–31], jet stirred reactor [32,33] and flow reactor [34,35] to provide more information on the NH_3 oxidation and NO formation pathways. Our understanding of NH_3 combustion and NO formation kinetics has advanced significantly thanks to these studies.

It is noted that experimental validation of NH_3 kinetic mechanisms against species profiles in NH_3 -fuelled laminar flames is scarce. This is in contrast to the large amounts of speciation data available for hydrocarbon flames which has proven to be essential in understanding the corresponding combustion chemistries [36]. This difference has to do with the intensive use of hydrocarbon fuels for decades; interests in NH_3 as the major fuel arise only recently. Unfortunately, significant discrepancies are present in flame NO predictions with existing NH_3 mechanisms, suggesting an urgent need for more flame experiments. For instance, Tang et al. [37] showed that for the six different kinetic mechanisms recently proposed for NH_3 combustion [32,38–42], differences exceeding 100 % can be observed in their NO predictions in a counterflow diffusion flame of NH_3/H_2 mixture. In another study, Brackmann et al. [43] showed that all the four tested kinetic mechanisms significantly overpredicted the peak NO mole fractions in a premixed NH_3 flame, with modelled results ranging from 200 % to 600 % higher than experimental data.

For a major portion of previous experimental research, NO concentration in the exhaust gas of ammonia flames was quantified with the primary aim of assessing the influences of different combustion techniques/parameters on NO emissions [44]. The measurements were typically performed with commercially available NOx analyzers based either on electrochemical sensors, Fourier transform infrared sensors (FTIR) or chemiluminescence detection of excited NO_2 (CLD). A relatively large sampling probe with typical sample flow rate of several liters per minute [45] was often required for such commercial equipment so that it is challenging to measure inside the thin reaction zone with sufficient spatial resolution. Often times, NO measurements were performed only in the post-flame exhaust region without characterizing flame thermochemical structures. For instance, Ramos et al. [46] measured the NO concentration in the exhaust of premixed NH_3/CH_4 /air flames using a water-cooled stainless steel probe with an inner diameter (*i.d.*) of 1.3 mm connected to a commercial CLD detector. They observed that NO emissions varied non-monotonically with NH_3 ratio in the NH_3 - CH_4 fuel mixtures; meanwhile they also noticed NO emissions were consistently overpredicted by three different mainstream kinetic mechanisms. Hayakawa et al. [47] characterized the exhaust gas of a strained NH_3 /air premixed flames between the reactant outlet and a downstream stagnation plate. Sampling channels (*i.d.* = 4.2 mm) for the product gas were driven in a circular pattern directly in the stagnation plate which were connected to a commercial FTIR gas

analyzer. The authors measured a peak NO emission of 3500 ppm under fuel-lean condition with an equivalence ratio $\varphi = 0.9$; they further showed a continuous decrease of NO accompanied with an increase of residue NH_3 emission with the further increase of φ . Similar variation trend with φ was also observed by Yu et al. [11] who recently measured the emissions of a conical-shaped premixed flame of $\text{NH}_3/\text{DME}/\text{air}$ using an industrial electrochemical NO sensor. Besides these three examples, there are indeed a large body of studies that report NO emissions in a variety of NH_3 combustion conditions, as summarized in a recent review [24]. It is noted that although these studies provide valuable information on emission characteristics of ammonia flames, they are probably not best suited for comprehensive model validation.

To be more useful for mechanism validation, NO measurements should ideally be performed in highly-controlled ammonia flames 1) with sufficient spatial resolution and 2) across a wide range of flame conditions. Such data can be obtained from non-intrusive *in-situ* optical measurements such as laser induced fluorescence (LIF) [37,43,48,49], Raman scattering [50], and light absorption spectroscopy [51]. Probe-based intrusive methods with subsequent chemical analysis such as MBMS [52,53] and FTIR [47] are also possible. Nevertheless, as mentioned, very few studies provided spatially-resolved data of NO in fundamental ammonia flames designed for kinetic model validation. Brackmann et al. [43] performed laser induced fluorescence (LIF) to measure $\text{NH}/\text{OH}/\text{NO}$ distributions in premixed stretch-stabilized ammonia-air flames. With the aid of the experimental temperature profiles from CARS measurement, quantification of the NO fluorescence signal to mole fraction were achieved through a bi-directional LIF approach, avoiding the need to consider the unknown collisional quenching. This work is highly relevant to model validation although measurements were limited to three flames that differ in equivalence ratios. Most recently, El Baba et al. [54] used LIF to measure axial profiles of NO and OH in low pressure $\text{NH}_3/\text{O}_2/\text{N}_2$ premixed flames. Collisional quenching was overcome as the signal was detected at its temporal peak, largely avoiding the influence of the subsequent LIF decay [55]. Quantification was achieved through calibration using either separate absorption measurement (for OH) or the standard addition method (for NO). In addition to providing valuable quantitative data across three different equivalence ratios, the authors also noted that none of the tested mechanisms could reproduce the measured NO consumption in the post-flame zone of a high temperature fuel-lean flame, thereby calling for further measurement of key intermediate species. Multi-pass Raman scattering method was tested by Zubairova et al. [50] to measure axial profile NO in a flat premixed ammonia-air flame with a lower detection limit of around 700 ppm. Yang et al. [51] used UV absorption spectroscopy to measure NO mole fraction profiles in premixed NH_3 - CH_4 flames with a maximum NH_3 fuel ratio of 50 %. Maclean and Wagner [56] and Bian et al. [57] performed among the earliest probe-based mass spectrometry measurement in low pressure NH_3/O_2 premixed flames, and these datasets were later shown particularly valuable for NH_3 reaction model development. Tian et al. [58] (Dyynslaegher et al. [53]) later performed chemical speciation in low-pressure $\text{NH}_3/\text{CH}_4(\text{H}_2)/\text{O}_2/\text{Ar}$ flames using MBMS. More recently, Osipova et al. resolved the chemical structures of NH_3 - H_2 flames at atmospheric [59] and at elevated pressures [52] with MBMS, with later work showing higher pressure conditions are beneficial for NO reduction. Spatially-resolved LIF NO measurements have also been performed in non-premixed flames. Tang et al. [37] reports axial profiles of NO in a series of counterflow diffusion flames with various NH_3/H_2 ratios and strain rates. Nevertheless, these studies are more targeted to hydrogen-enriched ammonia flames instead of neat ammonia flames. It may also be relevant to mention before the present ammonia research boom (2015-present), there were previous studies that targeted ammonia chemistry as reviewed by Miller and Bowman [60] and Glarborg et al. [61]. However, in contrast to the present interests of using ammonia as a carbon-free fuel, many of those earlier studies were motivated to understand the thermal De-NOx processes or to use

ammonia as a surrogate to study the evolution of fuel-nitrogen in biomass or coal combustion.

The scarcity of NO profile data in NH₃ flames and the necessity for further measurements are well recognized by authors of very recent studies [37,54], especially for pure NH₃ premixed flames in relation to NO formation. Apart from NO, emission of N₂O from NH₃ flames is also a major concern. It has been shown that the global warming potential of N₂O can be ~300 times higher than that of CO₂, making it essential to understand and control N₂O formation from NH₃ flames.

In this work, we aim to complement the few existing datasets by performing a comprehensive investigation on NO and N₂O formation characteristics in NH₃-fueled laminar premixed flames across a wide range of operating conditions. Motivated by the practical interests of NH₃-natural gas cofiring for industrial decarbonization, we also cover neat CH₄ flames and NH₃-CH₄ co-fired flames with varying fuel ratios. Spatially-resolved temperature profiles were obtained non-intrusively with tunable diode laser absorption spectroscopy (TDLAS) that targeted flame-generated H₂O molecules. NO and N₂O mole fractions were measured with spatial resolution using a novel in-house developed calibration-free methods combining micro-probe sampling and quantification with mid-infrared laser absorption spectroscopy (LAS). The experimental data was compared with results from corresponding numerical simulations using literature-based kinetic mechanisms for a critical assessment of their predictability. Kinetic analysis was also performed to shed light on how the various combustion parameters such as temperature, equivalence ratio, NH₃ mixing ratio and oxidizer composition may influence NO/N₂O evolution.

It may be worthwhile to mention that the present micro-probe sampling followed by quantification via LAS is straightforward and economic to implement, thereby offering a convenient and perhaps more accessible alternative to the advanced LIF / Raman / MBMS based NO measurement methods which require more intensive equipment investment and expertise.

The paper is organized as follows. Following this general introduction, the experimental platform, flame conditions and the novel calibration-free NO/N₂O measurement methods are described. Special attention is given to a number of measurement subtleties that are related to data reliability. Experimental results from the parametric study are then presented and discussed. Corresponding numerical simulations were also performed, and the results are used to explain, when appropriate, the observed experimental trends and to assess the accuracy of popular literature-based mechanisms.

2. Methodology

2.1. Burner setup and flame conditions

All the tested flames were stabilized on a standard water-cooled porous plug burner (Holthuis & Associates McKenna burner with a 60 mm burner plug diameter). Commercially available reactants of CH₄ (99.95 %), NH₃ (99.99 %), O₂ (99.99 %) and nitrogen (99.999 %) were precisely metered by thermal-based mass flow controllers (MFCs, Sevenstar D07, ±1 % accuracy) before being mixed and fed into the stainless-steel burner plug. Before the experiments, all the MFC were calibrated using high-precision digital bubble meters (Mesa Laboratories, Defender 510). A co-flow of N₂ gas was supplied to minimize interferences from ambient air flow and a stagnation plate was placed 20 mm above the burner surface to suppress the flickering the post-flame gases. The stagnation plate can at the same time serve as a steady boundary condition which is beneficial from a numerical simulation point of view. The burner was mounted on a motorized linear translation stage (Yiheng, TDPF200) to facilitate spatially resolved measurements. A schematic is given in Fig. 1 along with direct images of representative neat CH₄ and neat NH₃ flames.

A series of flames with varying fuel NH₃ fraction (X_{NH_3}), equivalence ratio (φ), oxygen fraction (X_{O_2}) and reactant flow velocities at the burner

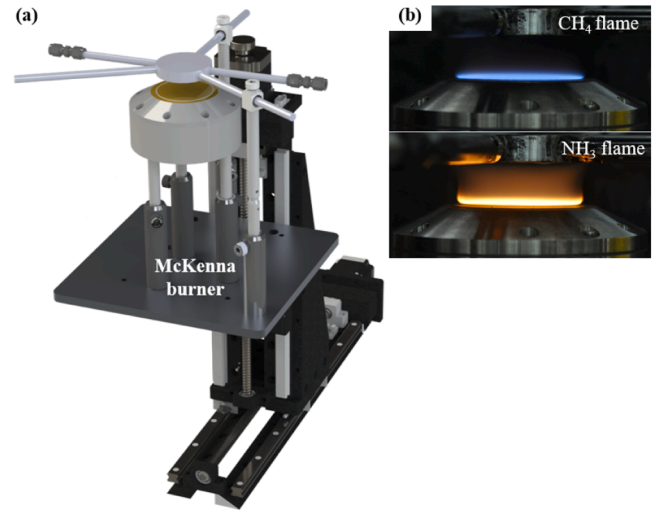
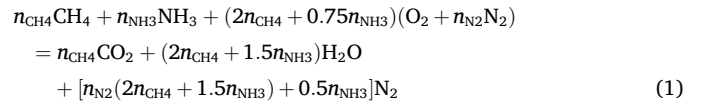


Fig. 1. Schematic of the burner assembly (a); also included is photo image of a representative neat CH₄ flame and neat NH₃ flame (b).

exit (V) were established. Specific definitions of these parameters are given below, with reference to the following reaction equation for complete NH₃-CH₄ combustion:



where n is the number of moles for each corresponding reactant. We define fuel NH₃ fraction (X_{NH_3}) as the molar fraction of NH₃ in the fuel mixture:

$$X_{\text{NH}_3} = \frac{n_{\text{NH}_3}}{n_{\text{NH}_3} + n_{\text{CH}_4}} \times 100\% \quad (2)$$

and oxygen fraction (X_{O_2}) as the mole fraction of oxygen in the oxidizer which is a mixture of O₂ and N₂:

$$X_{\text{O}_2} = \frac{1}{1 + n_{\text{N}_2}} \times 100\% \quad (3)$$

The equivalence ratio (φ) is the actual fuel to oxidizer mass ratio divided by the stoichiometric fuel to oxidizer mass ratio, so the flame is fuel-rich when $\varphi > 1$ and fuel lean when $\varphi < 1$:

$$\varphi = \frac{\text{fuel} - \text{to} - \text{oxidizer ratio}}{(\text{fuel} - \text{to} - \text{oxidizer ratio})_{\text{st}}} \quad (4)$$

Three series of flames were measured with detailed flame conditions summarized in Table 1. In the N series flame, X_{NH_3} was varied from 0 (neat CH₄ flame) to 1 (neat NH₃ flame) with an interval (Δ) of 0.1, while φ and V were fixed at 0.9 and 5 cm/s, respectively. For the V series flame, the burner exit velocity (V) was varied with X_{NH_3} and φ fixed. A change of reactant flow velocity (V) would obviously change the thermal power of flame; however, it should be noted that our primary motivation to vary V is to change the flame temperature. In the E series flame, φ was

Table 1
Detailed flame conditions studied in this work.

Flame Series	Ammonia fraction (X_{NH_3})	Equivalent ratio (φ)	Oxygen concentration (X_{O_2})	Burner exit velocity V / (cm·s ⁻¹)
N	0~1($\Delta=0.1$)	0.9	0.21/0.30/0.35	5
V	0.2	0.9	0.21/0.30/0.35	5~8($\Delta=1$)
E	0.2/0.4/0.6/1.0	0.7~1.3 ($\Delta=0.1$)	0.21/0.30/0.35	5

varied from 0.7 to 1.3 to establish both fuel-lean to fuel-rich flame conditions. Note X_{O_2} is always a secondary parameter in each of the three series. Finally, we give two naming conventions for reference to a specific flame condition: $N_{X_{NH_3}0.1_{X_{O_2}0.3}}$ and $E_{\varphi0.8_{X_{NH_3}0.2_{X_{O_2}0.21}}$, with self-explanatory meanings.

2.2. Temperature measurements

We utilized a mid-infrared laser with emissions near $2.5 \mu m$ to target the absorption spectral lines of H_2O at 4029.524 cm^{-1} and 4030.727 cm^{-1} to measure temperature. The measurement principles have been given in previous studies [62,63]. A schematic of the setup is given in Fig. 2 where the probing mid-infrared laser light was seen to pass through the flame before being detected by the photodetector (PD). To ensure sufficient spatial resolution in the flame axial direction, a series of lens was used to focus the beam to $150 \mu m$ at the flame center ($210 \mu m$ at the flame edge). Measurements were made at different HAB by moving the burner assembly (translation stage not shown) relative to the fixed laser beam.

The present flame, although being flat in the core region, does contain radial non-uniformity at the edge where the cold nitrogen curtain is present. LAS is inherently a line-of-sight method so that such effect of non-uniformity on measurement accuracy needs to be considered. This is achieved in our work through careful selection of absorption line pairs. In particular, by selecting transition lines centering at 4029.524 cm^{-1} and 4030.727 cm^{-1} that feature a large lower state energy ($>2000 \text{ cm}^{-1}$), it was shown that the line-of-sight temperature measurement agrees with the one obtained through tomographic reconstruction within an accuracy of 3.5 % in the plateau flame core region. Interested readers are referred to our previous work for details [64].

2.3. NO/ N_2O measurements

As mentioned in the introduction section, spatially-resolved measurements of NO/ N_2O in NH_3 flames are not trivial. Here we combine micro-probe sampling with calibration-free quantification of the sampled gas using an in-house developed mid-infrared LAS setup. Special efforts were made 1) to minimize the intrusion of the sampling probe on the flame; 2) to ensure the sampled gas does not alter in composition during the sampling process and 3) to have accurate quantification of NO/ N_2O in the sampled gas over a large dynamic ratio without the need of pre-calibration. These efforts will be elaborated soon, while it may be worthwhile to emphasize that the feature of calibration-free is essential. Since we cover a wide range of flame

conditions, the composition of the flame gas may vary significantly. In this regard, NO/ N_2O concentrations need to be quantified without a *prior* knowledge of the bath gas composition; this makes pre-calibration impractical.

2.3.1. Experimental setup

A schematic of the measurement setup is given in Fig. 3. A ceramic probe (Kyocera Corporation) with an inner diameter of $100 \mu m$ is inserted into the flame to withdraw the flame gases. The probe is connected through a gas-tight straight fitting (with a graphite sealing gasket) to a stainless-steel tube before being directed to the inlet port of the gas cell. A vacuum pump connecting the exit port of the cell provides the necessary pressure difference to drive the sampled gas. To prevent water condensation during the sampling process, the entire sampling pipeline and the gas cell are heated electrically and thermally insulated. Note, since the temperature and pressure of the sample in the gas cell needs to be accurately known for subsequent concentration quantification, their accurate control and measurement is required. In this regard, all the heating equipment is precisely regulated by PID controllers to maintain a temperature at $393 \pm 1 \text{ K}$. During the actual measurement, the high-temperature gas cell was firstly pumped into the vacuum state; the flame gas was then sucked into the gas cell whose absolute pressure was monitored real-time with a pressure transducer. Once the target pressure was achieved (typically at 30 kPa absolute, rationale provided later), both the inlet and outlet ports of the cell were closed so that the sampled gas was sealed in the cell with a pre-known temperature and pressure. Its NO and N_2O concentrations were then determined by the TDLAS technique. Note that a single path length of the 35 cm gas cell was used for CH_4 - NH_3 cofired cases, while the effective absorption path was extended to 105 cm (3 passes) for the neat CH_4 cases to increase sensitivity. The reason for our choosing *ex-situ* over *in-situ* flame TDLAS NO measurement has to do with detection limit: The line-strength of NO transition decreased significantly as temperature increased from ambient to flame temperature, leading to a limited lower detection limit if *in-situ* measurement was attempted. We estimated that for the present flame the signal-noise ratio would become unacceptable when $NO < 200 \text{ ppm}$; this is insufficient for measuring flames with low NH_3 proportions. Additionally, the strong sensitivity of NO line-strength to temperature also poses challenges to account for the non-uniform radial distribution of temperature, especially at the flame edge where temperature drops significantly due to mixing with curtain N_2 .

Before detailing the optical setup, we would like to mention that it is not possible to perform measurements with the sampled gas continuously flowing through the cell. This is because a microprobe with a tiny sampling tip was used to maximize spatial resolution and to avoid flame

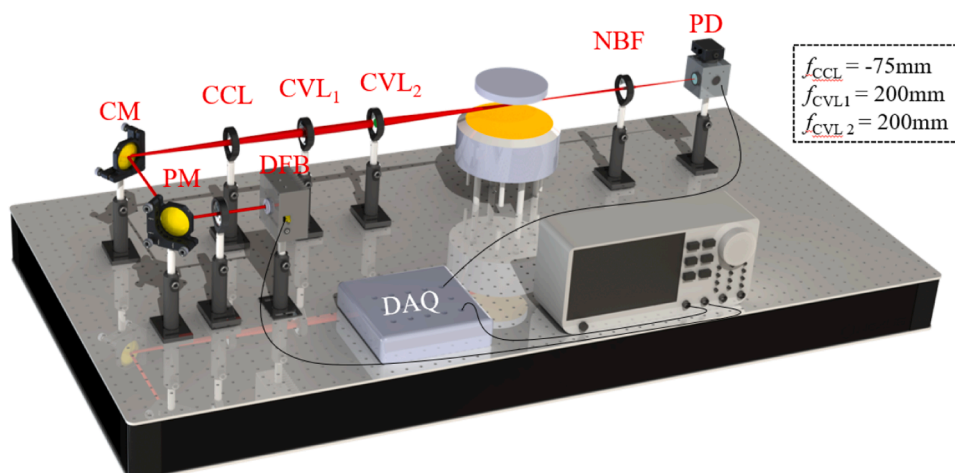


Fig. 2. Schematic diagram of temperature LAS measurement system: DFB, distributed feedback laser; PD, photodetector; Iris, diaphragm; CVL, convex lens; CCL, concave lens, NBF, narrow-band filter; PM, plane mirror; CM, concave mirror; DAQ, data acquisition module.

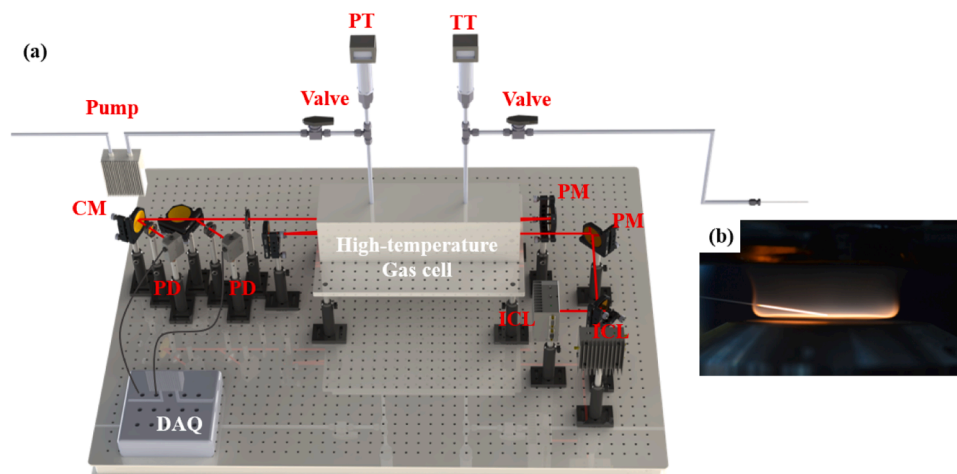


Fig. 3. Schematic of the burner assembly along with the NO/N₂O measurement setups (a) and direct image of a representative NH₃/CH₄ flame (b). ICL: interband cascade laser; PM: plane mirror; CM: concave mirror; PD: photodetector; PT: pressure transducer; TT: temperature transducer; DAQ, data acquisition module.

disturbances; the accompanied consequence is that the sample flow rate is order of magnitude lower than required in typical commercial NO_x gas analyzers. In this regard, the aforementioned procedure is necessary to accumulate a sufficient amount of sample gas for accurate concentration determination. However, one drawback of the procedure is that it can take tens of minutes to fill the cell to the required pressure; this poses challenges to both the structural reliability of the microprobe (as it needs to be inserted in the high-temperature flame) and the performance stability of the sampling system (no leakage of ambient air inside the vacuum gas cell and chemical stability of the sampled gas need to be ensured). Of course, one could minimize the volume of the gas cell to reduce the required amount of the sample gas and thus the sampling time; however, a sufficient beam path (L) is necessary to have an acceptable SNR and lower detection limit of the laser absorption measurement. A compromise we finally made was to choose a gas cell with $L = 35$ cm and an internal volume of 650 mL. In the worst-case scenario when the probe was inserted into the maximum temperature region of the flame (lowest density), the sampling time was approximately 4 minutes.

NO/N₂O mole fractions in the gas sample were quantified via mid-infrared LAS. As can be seen in Fig. 3, the collimated emission from an interband cascade laser (ICL) was directed by a planar mirror (PM, $D=25.4$ mm) to pass through the gas cell before being refocused by a concave mirror (Thorlabs, CM254-100-M01, $D=25.4$ mm) onto a MCT photodetector (VIGO system, PVI-3TE-6). A narrow bandpass filter (Favision, BP-5230 $\pm$ 50 nm or BP-4600 $\pm$ 45 nm, depending on the measurement target) was placed before the photodetector to reduce the emissions from sources other than the laser. A multi-channel data acquisition module (National Instrument, USB 6363) was used to perform the signal trigger synchronization and data acquisition (200 kS/s sampling rate for DAS and 2MS/s sampling rate for WMS). The absorption spectral lines were near 1900.075 cm⁻¹ for NO and near 2191.421 cm⁻¹ for N₂O. Such specific transitions were chosen after considering the need to minimize spectral interference from major combustion products such as CO, CO₂ and H₂O.

As both neat CH₄ and neat NH₃ flames were included in this work, NO/N₂O mole fractions can vary by several orders of magnitudes across different flame conditions, necessitating a large measurement dynamic ratio. This is realized by combining the direct absorption spectroscopy (DAS) technique for high concentration range and the wavelength modulation spectroscopy (WMS) technique for low concentration range. Note that both the techniques are calibration-free, they also share the same optical setup so that no physical movement of any component is needed when converting. The main differences lie in modulation of the laser injection current and the spectral fitting during the data post-

processing stage. An introduction to the DAS measurement for NO has been given in a previous work [65], and a brief description of the methodology for calibration-free WMS measurement of NO and N₂O is given below.

In calibration-free WMS, the characterized laser tuning optical frequency under conditions of slow scan and high-frequency modulation, along with the spectroscopic parameters of the target species, lineshape function, and the measured incident laser intensity I_0 are required to create spectroscopic models that fit the transmitted laser intensity. Both the modelled and the measured transmitted laser intensity are demodulated using the same software-based lock-in amplifier to extract the first harmonic (1f) and second harmonic (2f) signals. Notably, the first-harmonic (1f) normalized second-harmonic (2f) signal (WMS-2f/1f) is selected as the fitting target due to its robustness in harsh environments. Once the minimum sum-of-squared error (SSE) is achieved during the fitting process, the gas concentration of the target species can be quantitatively determined. A more detailed explanation of the working principles and implementation of the calibration-free WMS strategy is available in a previous study on NH₃/CO measurements [66] as well as in [67].

2.3.2. Sensor validation

We next validate the performance of our TDLAS sensor by measuring gas mixtures with pre-known NO/N₂O mole fractions. We procured standard gas cylinders of NO/N₂O mole fractions of 10, 100, 1000 and 4000 ppm balanced by N₂. Mixtures of other NO (N₂O) concentrations in the range between 2–4000 ppm (2–500 ppm) were prepared with an in-house gas divider. The test gas was then directed to the reactant inlet of the McKenna burner to flow through the porous plug. The microprobe placed 2.0 mm above the burner then withdrew the test gas to the gas cell until a desired cell pressure was reached. This process was exactly the same as that followed in actual flame experiment as described earlier.

The validation results are given in Fig. 4 for both NO (a) and N₂O (b), where WMS was used for nominal concentrations below 200 ppm and DAS applied for above (the pink area shows a comparison of the measurement results using both DAS and WMS at this concentration, which served as a cross-validation for both methods). Note that values in the y-axis were derived by fitting the measured DAS/WMS spectra (examples given in Fig. 4(c)/(d)) without the need for pre-calibration. Different gas cell pressures were tested with essentially the same results, further demonstrating the satisfactory performance of our calibration-free spectra fitting approach. The maximum relative error of the measurement was within 4 %, originating mainly from uncertainties in spectroscopic parameters and temperature and pressure of the gas cell.

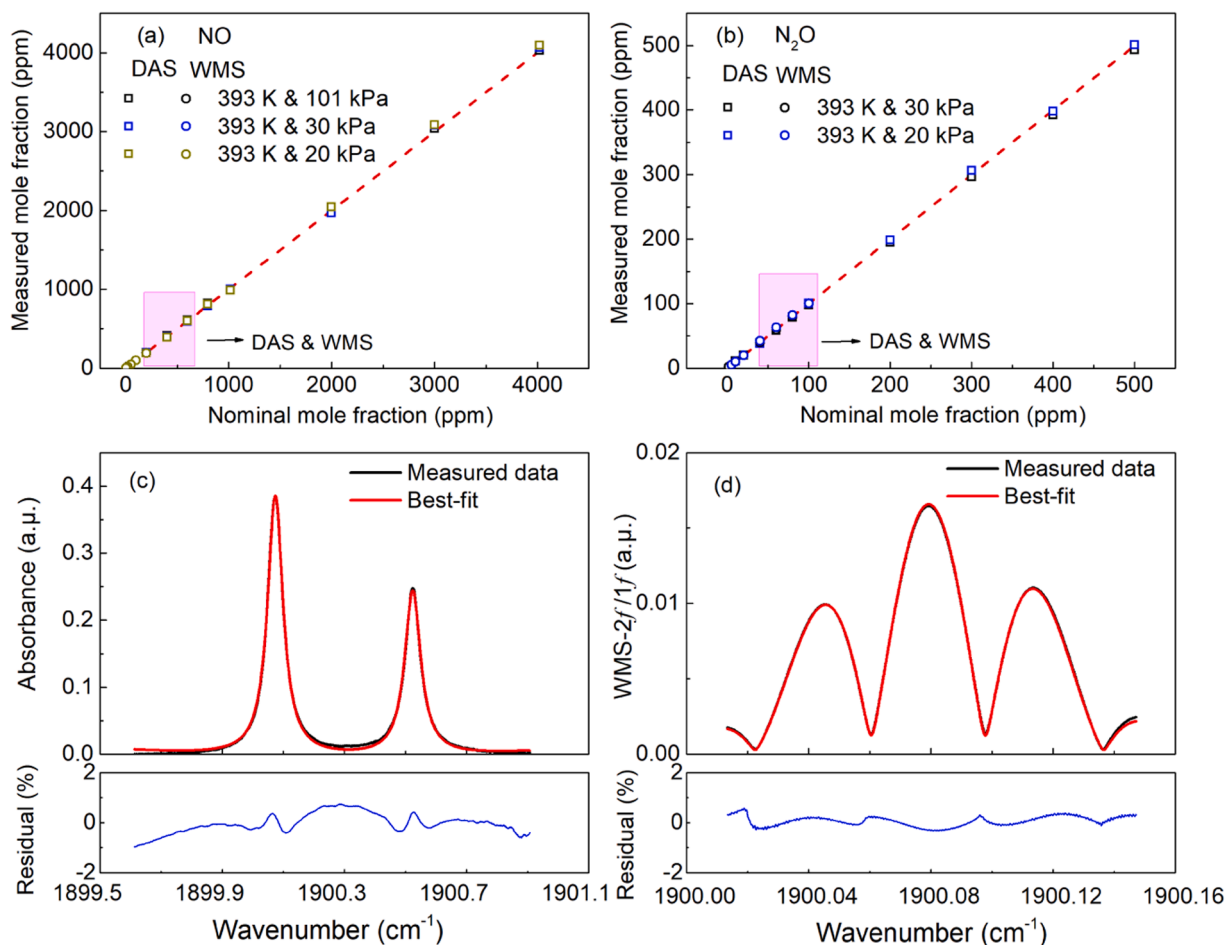


Fig. 4. Measured NO (a) and N₂O (b) mole fractions for gas mixtures with pre-known compositions and typical DAS/WMS spectra (c)/(d).

The above results confirmed the accuracy of the present TDLAS NO/N₂O sensor, resolving the third point mentioned in the beginning of this subsection. It also helped to confirm the sampling system to be free of leakage, because otherwise the leaked air would have altered the concentration of the target gas through dilution. However, it should be noted that the test gas was composed of only NO/N₂O and N₂, while in actual experiments the flame gases may contain O₂ which may react with NO to form NO₂ during the sampling process. Such reactions, if proceeded in a significant rate, would alter the NO concentration and lead to biased measurement results. It is thus imperative to assess this undesirable effect in a quantitative manner, especially when considering the sampling process could last for a significant amount of time due to the small probe size. In this regard, we performed NO oxidation experiments in the gas cell with real time monitoring of NO mole fractions for a time period of 10 minutes. The parameters we swept include NO and O₂ initial mole fractions (balanced by N₂) as well as the temperature of the gas cell, and the results are shown in Fig. 5. We note from Fig. 5(a) and (b) that, as expected, the reaction rate for NO + ½ O₂ = NO₂ increased with the increase of the initial concentration of either reactant. Interestingly, Fig. 5(c) indicated that the reaction rate decreased with the increase of temperature, exhibiting a negative global activation energy. This suggests that maintaining an elevated gas cell temperature is beneficial in keeping the compositional fidelity of the sampled gas. As a matter of fact, a reaction rate constant for the global reaction of NO + ½ O₂ = NO₂ can be derived from the measured data and the result agreed well with literature data. We remark that the global rate constant was pressure-independent so that reaction rate would also decrease with the decrease of gas cell pressure. In this regard, keeping the gas cell at low pressure is also beneficial although compromise has to be made due to

the decreased measurement sensitivity of the TDLAS sensor. We finally show in Fig. 5(d) NO evolution with time for a scenario with NO mole fraction of 5000 ppm and O₂ mole fraction of 10 %, corresponding to a most challenging case for all the flames we tested. The cell temperature and final pressure were set at the same value as we chose for the actual flame experiments at 393 K and 30 kPa, respectively. Note that the cell pressure was increasing from 0 to 30 kPa as the sampled gas continued to fill the cell (the cell was initially vacuumed). Even for this critical case, NO reduction at the end of the 5-minute period (typical sampling time) was seen to be only 0.4 %, eliminating the concerns for compositional fidelity due to NO oxidation. This result helps to resolve the second point mentioned in the beginning of this subsection.

2.3.3. Effects of probe intrusiveness in flames

A remaining concern is the intrusiveness of the sampling process, i.e., whether the present microprobe can withdraw flame gases without distorting flame chemistries. Note that the performance of microprobe sampling has been well demonstrated in the literature on hydrocarbon flames [68,69]; in particular, Jiang et al. [70] and Li et al. [71] compared probed-measured species concentration data with non-intrusive optical method and reported consistent results from both methods. Here, we intend to provide further evidence for NH₃ flames. It is relevant to mention that our main target is NO and N₂O whose spatial distribution is usually centered around the peak temperature region. Therefore, measurement accuracy near the reaction front is critical; this is notably different from our previous work focusing on soot precursors [72] which are abundant mainly in the relatively low temperature fuel pyrolysis zone. We chose to set up a counterflow flame of CH₄/O₂/Ar flame with doped NO to assess the intrusiveness of the probe.

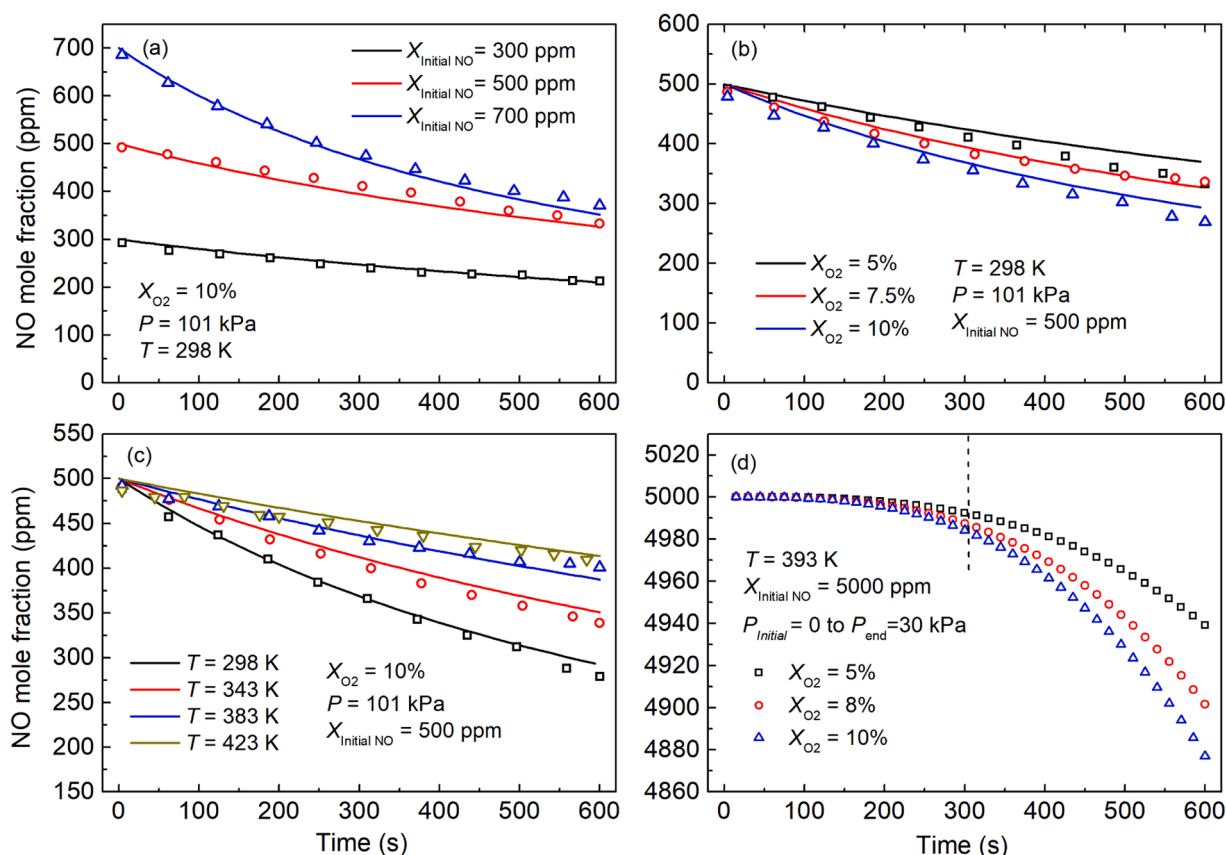


Fig. 5. Effects of initial NO (a), O_2 mole fraction (b), and temperature (c) on the temporal evolution of NO in the gas cell; (d) NO oxidation at 393 K for an NO mole fraction of 5000 ppm and O_2 mole fraction of $5, 8$ and 10% , final cell pressure 30 kPa .

Counterflow flame is more sensitive to probe intrusiveness compared to burner-stabilized flat flame since the former has even a smaller flame area and steeper species gradient [73]. In this regard, our test is more critical and can serve as a validation for future work in the more challenging counterflow flame. We avoided the presence of N_2 in the flame so that NO in the flame can only come from the NO-doped Ar gas. The benefits of this arrangement originate from the minimization of uncertainty in NO prediction so that the level of agreement between the experimental data and the numerical prediction directly reflects the fidelity of the probe sampling process.

The counterflow burner was described in a previous work [70], it has two aerodynamically contoured nozzles (*i.d.* 10 mm) arranged to oppose each other with a separation distance of 8 mm . The fuel side was 50% CH_4 and 50% Ar (doped with 1060 ppm NO) and the oxidizer side was 21% O_2 balanced with neat Ar. Both the fuel and oxidizer stream have a nozzle exit velocity of 25 cm/s , and a curtain flow of Ar was used to shield the flame from ambient air. Probe sampling was performed along the center axis of the flame. Quantitative comparison between the measured data and numerical simulation is given in Fig. 6 where a satisfactory agreement can be seen. The simulation was performed with the Opposed Flow Module of the Chemkin Pro package using a well validated mechanism for CH_4 fuel. As mentioned, since the uncertainty in the mechanism describing NO formation is avoided, the simulation is highly reliable (the two mechanisms predict the same results) and such level of agreement directly confirms the validity of our present sampling methods.

Our primary goal was to provide spatially resolved NO/ N_2O data, although the NO data shown in Fig. 6 demonstrates the performance of our method in terms of spatial resolution, we note during the course of the work that N_2O profile near the reaction zone can exhibit even stronger axial gradient. To make sure such gradient can be resolved, we

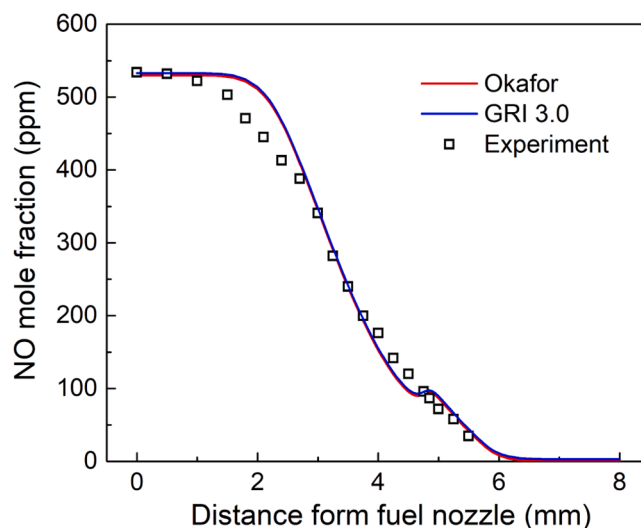


Fig. 6. Comparison between experimental and numerical predicted axial profiles of NO mole fraction in a counterflow flame of $CH_4/O_2/Ar$ with doped NO. The numerical simulations were performed with two different reaction mechanisms of Okafor and GRI 3.0 Mech.

performed additional tests using microprobes with a range of inner diameters. The results in Fig. 7(a) shows the measured N_2O mole fraction in a NH_3-CH_4 burner stabilized flame with $X_{NH_3} = 0.2$, $\phi = 0.7$ and $V = 5 \text{ cm/s}$. Numerical results are shown to compare with the experimental data obtained with three different probes with inner diameters of $0.1, 0.15$ and 0.2 mm , where it can be noticed that N_2O exhibited a narrow

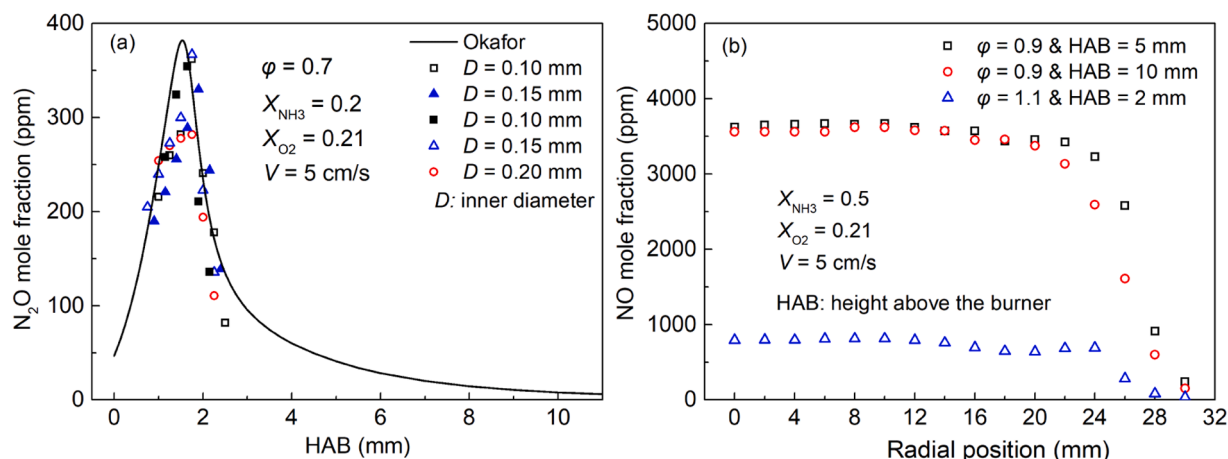


Fig. 7. Comparison of axial profiles of N_2O as measured by probes with different inner diameters (a) and measured radial profiles of NO in burner stabilized premixed flames (b). Note that numerical prediction using the Okafor mechanism is also given in (a) and that detailed flame conditions are given in the figure legends.

peak with FWHM less than 1.8 mm. Data obtained with the largest probe of 0.2 mm was seen to be insufficient to resolve the narrow profile, exhibiting strong smearing effects with the measured peak notably lower than those measured with the smaller 0.1 and 0.15 mm *i.d.* probes. On the other hand, the smaller probes showed comparable and repeatable performances, suggesting a probe of 0.15 mm *i.d.* was sufficiently small. In terms of radial variation, Fig. 7(b) presents the measurement scan along the radial direction at a specific axial position. As expected, the radial gradient was minimal unless at the edge of flame, posing no issues to the present sampling method.

2.4. Numerical modelling

Unless stated otherwise, all the flames in this work were simulated using the 1-dimensional solver of the premixed burner stabilized stagnation flame module in the Chemkin package. Boundary conditions of reactant composition, temperature and flow velocity were supplied at the burner surface and the downstream stagnation plate was treated as wall boundary with a fixed temperature. In particular, the temperatures of the burner surface and the stagnation plate (both of which are water-cooled) were set at room temperature and 500 K, respectively. Note, although no measurement was performed for these boundary temperatures, we have performed sensitivity analysis and confirmed the predicted temperature and $\text{NO}/\text{N}_2\text{O}$ profiles are not sensitive to the exact settings.

Adaptive gridding was applied, and the level of grid refinement was controlled by reducing the solution gradient and curvature below a predefined value of 0.1 and 0.5, respectively. Axial profiles of temperature were obtained by solving the energy equation with gas radiation from H_2O , CO_2 , CH_4 , CO and NO [74,75] using an optically thin approximation.

To assess and compare the performance of existing reaction mechanisms, we performed the simulation with 5 different mainstream kinetic schemes for $\text{NH}_3\text{-CH}_4$ combustion that are available in the literature.

These mechanisms include the Tian Mech [76], GRI Mech 3.0 [77], Konnov Mech [78], Okafor Mech [39] and the CEU Mech [27].

3. Results and discussions

3.1. Flame structure

Direct photographs of a series of $\text{NH}_3\text{-CH}_4$ fuel-lean ($\phi = 0.9$) flames are shown in Fig. 8 with an increasing NH_3 fraction X_{NH_3} . As can be seen, the flames are fairly flat except near the edge, where a slight upward curving can be observed due to the surrounding shield N_2 flow. The color of the reaction front changes from blue for the neat CH_4 flame ($X_{\text{NH}_3} = 0$) to orange with increasing saturation with the increase of X_{NH_3} . The blue color originates from the chemiluminescence of CH^* radicals and the orange one from NO_2^* and NH_2^* radicals that become increasingly abundant as more NH_3 is added [79,80].

We present in Fig. 9(a) and (b) a comparison of numerical and experimental results of $\text{NO}/\text{N}_2\text{O}$ and temperature for a series of flame conditions. As can be seen, for a specific flame with $X_{\text{NH}_3} = 0.2$, $\phi = 0.9$ and $V = 5$ cm/s, all the predictions captured the key feature of the temperature profile: *i.e.*, a rapid increase of temperature near the burner exit, the subsequent plateauing, and the final convergence toward the temperature of the stagnation plate. The Tian Mech predicted a slightly higher peak flame temperature while the others agreed fairly well with each other and with the experimental measurements. However, the discrepancy in predicted NO mole fraction among different mechanisms was seen to be significant. For instance, the peak NO mole fraction as predicted by the GRI mech was 4620 ppm, which is 1.7 times higher than that the value computed with the Okafor Mech. This indicates that notable uncertainties in the rates and/or kinetic routes related to fuel- NO_x formation exist. The experimental data also showed that N_2O formed rapidly, and its mole fraction peaked near the flame front at $\text{HAB} = 1.2$ mm and quickly decreased to 0 at $\text{HAB} = 2.5$ mm. Such a general profile was qualitatively captured by all the mechanisms, while

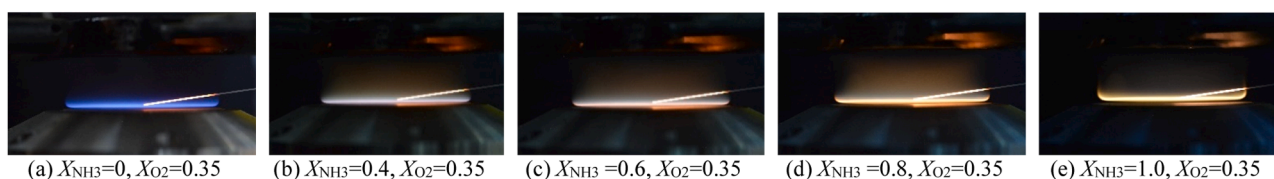


Fig. 8. Direct photographs of burner-stabilized $\text{NH}_3\text{-CH}_4$ flames with increasing X_{NH_3} at $\phi = 0.9$, $V = 5$ cm/s. Note the microprobe which was inserted in a tilted angle is also present.

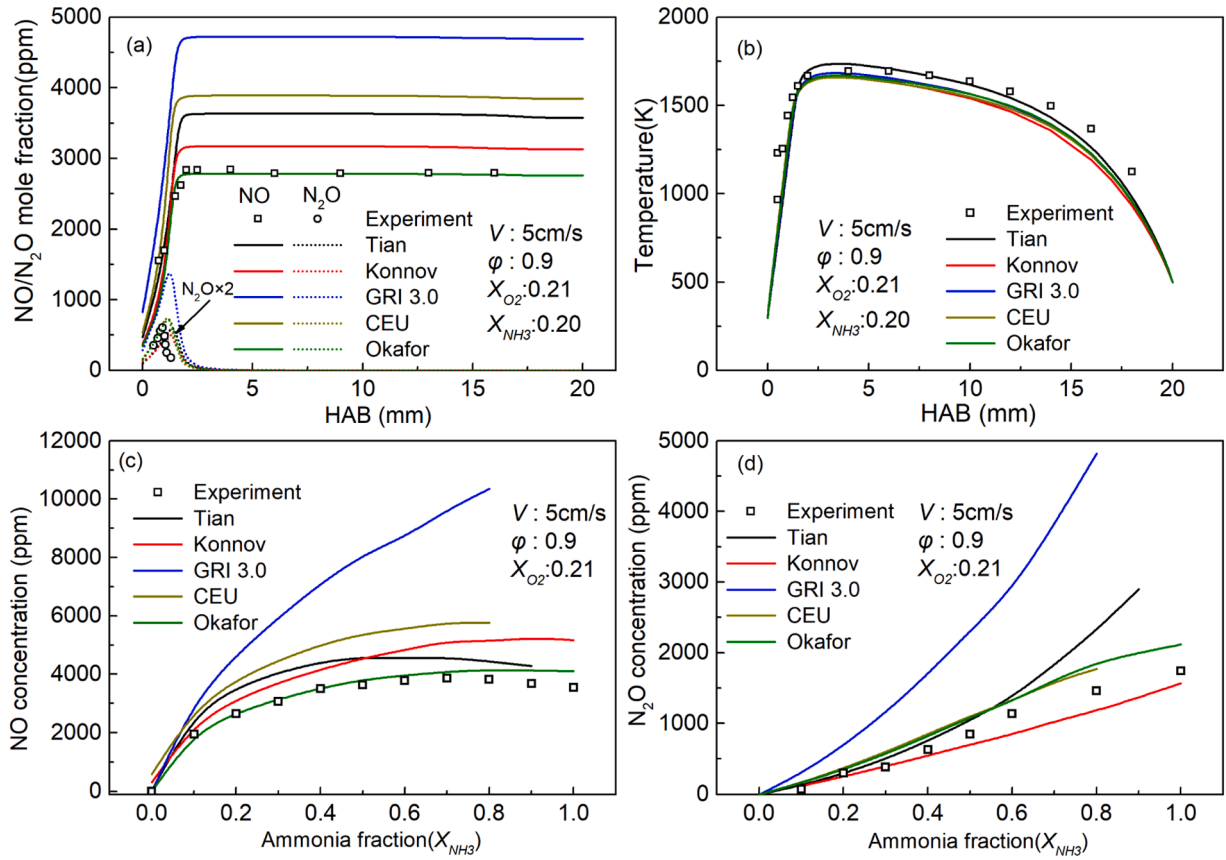


Fig. 9. Axial NO/N₂O mole fraction profiles (a) and temperature (b) in a NH₃-CH₄ cofired flame with $X_{NH_3} = 0.2$, $\phi = 0.9$ and $V = 5$ cm/s; variation of the peak NO (c) and N₂O mole fraction (d) with X_{NH_3} .

quantitatively the GRI Mech over-predicted and the Okafor Mech performed the best.

To further assess these mechanisms, we show in Fig. 9(c) and (d) respectively the peak NO and N₂O mole fraction in fuel-lean flames with X_{NH_3} spanning from 0 to 1.0. As can be seen from the experimental data (c), with the increase of X_{NH_3} , NO mole fraction firstly increases, peaking at $X_{NH_3} = 0.7$ and then decreases. The mechanism of this interesting non-monotonic behavior shall be explained shortly, but here it is important to notice the different trends as predicted by the various kinetic mechanisms. The numerical results calculated with the Okafor mechanism were seen to exhibit the best agreement with the

experimental data while the GRI mech was not able to capture the decrease of NO formation for $X_{NH_3} > 0.7$. As for peak N₂O mole fraction (d), the GRI mech was seen to give the largest discrepancy when compared with the experimental data and the Okafor Mech, the CEU Mech and the Konnov Mech were seen to have better agreement.

Since the present work is a rather comprehensive parametric study, it may not be manageable to perform kinetic analysis for all the flame cases with each of the mechanisms tested. As such, considering the demonstrated better performances of the Okafor Mech, in subsequent sections we chose to perform kinetic pathway and sensitivity analysis using only the Okafor Mech.

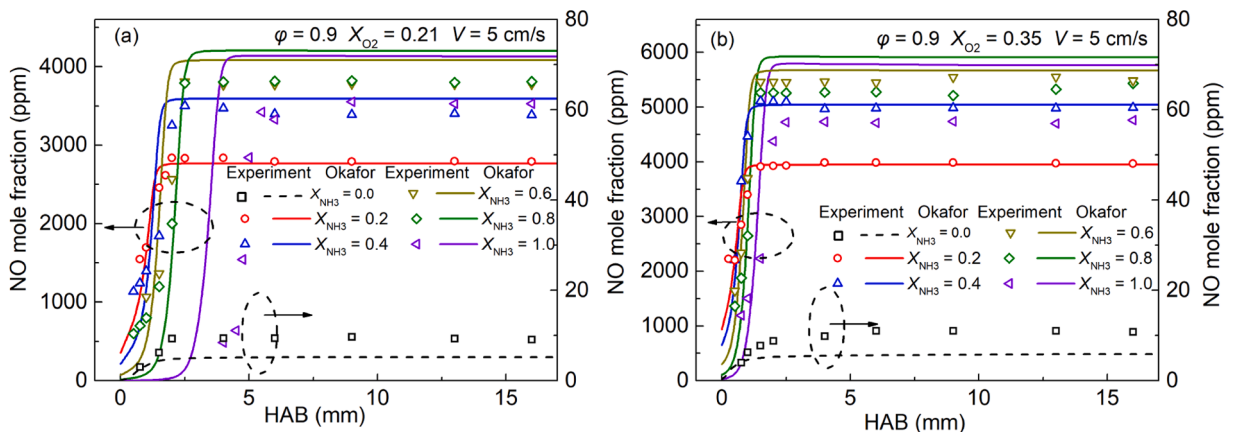


Fig. 10. Axial NO mole fraction profiles in fuel-lean ($\phi = 0.9$) N series flames for various X_{NH_3} with $X_{O_2} = 0.21$ (a) and 0.35 (b).

3.2. NO formation characteristics

3.2.1. Effects of NH_3 fraction (X_{NH_3}) on NO formation

We show in Fig. 10(a) and (b) the axial NO distribution for selected cases in the N flames series in which X_{NH_3} is the primary parameter to be varied. A consistent trend of a first rapid increase at small HABs followed by a plateauing in the post-flame zone can be observed for these fuel lean NH_3 -doped flames. In addition, the position at which NO reached its maximum point was continuously delayed as X_{NH_3} increased. This is consistent with the fact the flame front was also shifted downstream with the increase of X_{NH_3} . Note an increase in X_{NH_3} corresponds to reduction of reactivity and adiabatic flame speed. For the present flame series that shared the same reactant consumption speed (i.e., constant inlet velocity $V = 5 \text{ cm/s}$), an increased flame temperature is implied to counteract the reduced reactivity, this is in turn realized by shifting the flame further away from the burner surface so that heat loss to the burner can be reduced with the increase of X_{NH_3} . Such experimental trends were also well captured by the numerical simulation, except for the neat NH_3 flame of $\text{N}_{X_{\text{NH}_3}1.0}\text{O}_20.21$ where the experiment showed a peak NO location (HAB) larger than the experiment (Fig. 10(a)). It is noted during the experiment that for this neat NH_3 -air flame, the location of the flame front can be easily disturbed since this flame condition was not far from the blow-off conditions (i.e., V was close to corresponding adiabatic flame speed).

We show in Fig. 11 the measured and predicted peak NO mole fraction in N flame series with $X_{\text{O}_2} = 0.21, 0.30$ and 0.35 . In terms of the measured effects of X_{NH_3} on NO formation, three different regimes can be observed regardless of the value of X_{O_2} : Regime I with $0 \leq X_{\text{NH}_3} < 0.4$ where NO formation showed a rapid increase with X_{NH_3} . Regime II with $0.4 \leq X_{\text{NH}_3} < 0.7$ exhibited a moderate increase of NO with X_{NH_3} , and Regime III in which NO formation decreased as X_{NH_3} increased beyond 0.7 . Quantitatively, take the case with $X_{\text{O}_2} = 0.21$ (black curve) for example: the NO mole fraction increased from $\sim 10 \text{ ppm}$ at $X_{\text{NH}_3} = 0$ all the way to $\sim 3800 \text{ ppm}$ at $X_{\text{NH}_3} = 0.7$ before decreasing gradually towards $X_{\text{NH}_3} = 1.0$. Most notable is the fact that even a small amount of NH_3 addition ($X_{\text{NH}_3} = 0.1$) can result in an increase of NO formation by a significant amount.

NO in the neat CH_4 flame with $X_{\text{NH}_3} = 0$ can only be formed by taking N atoms from N_2 , while in NH_3 - CH_4 co-fired flames NO can also be formed from N-containing fuel molecules. We performed sensitivity and reaction pathway analysis to identify the detailed kinetic mechanisms that are responsible for the observed effects of X_{NH_3} . The results showed that in the present neat CH_4 flames, prompt [81] route is the major NO formation pathways. For the co-fired and neat NH_3 flames, on the other hand, all reactions that exhibit significant sensitivity coefficients include

NH or NH_2 radicals, confirming NO formation in the co-fired flames is largely related to the fuel NO route. The increasing importance of the fuel-NO route explains the initial rapid increase of NO as shown in Fig. 11.

Kinetic analysis was also performed to facilitate a more comprehensive understanding of the NO formation routes under different X_{NH_3} conditions. The results are illustrated in Fig. 12 for the co-fired flames with $X_{\text{NH}_3} = 0.2, 0.7$ and 0.9 . The red routes represent important reactions that exhibit positive sensitivity coefficients to NO formation while the green ones have negative sensitivity coefficients. The line thickness indicates the rate of production (ROP) of the corresponding reaction integrated over of the entire domain. The percentage values near each reaction path were obtained by normalizing the corresponding integrated ROP values by the initial hydrogen abstraction of NH_3 ($\text{NH}_3 + \text{O}/\text{OH}/\text{H} \rightarrow \text{NH}_2$) in the co-fired flame with $X_{\text{NH}_3} = 0.2$ (i.e., $\text{N}_{X_{\text{NH}_3}0.2}\text{O}_20.21$).

It is observed the major NH_3 conversion path was through $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N} \rightarrow \text{N}_2$, with $\text{NH}_i \rightarrow \text{NNH} \rightarrow \text{N}_2$ playing a secondary role. Only a minor portion of CO_2 from the CH_4 combustion would interact with NH to form HNO that finally leads to NO formation, while the major NO formation route in the co-fired case originate from the fuel-NO route of $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{HNO} \rightarrow \text{NO}$. Note HNO is a particularly important intermediate which is nearly 100 % converted to NO. A minor route is related to O radical through $\text{NH} + \text{O} = \text{NO} + \text{H}$. NO can also be reduced back to N_2 with the most contributing channel being $\text{NH}_i + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$. In addition, although to a much less extent, NO can also be converted to NO_2 through $\text{HO}_2 + \text{NO} = \text{NO}_2 + \text{OH}$ and N_2O through $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$, the latter of which is the primary N_2O formation channel to be detailed in a subsequent section.

A comparative analysis among the co-fired cases with different X_{NH_3} shows that with the increase of X_{NH_3} , ROP contributions from $\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$ and $\text{NH} + \text{OH} = \text{HNO} + \text{H}$ first rise and then fall. With $X_{\text{NH}_3} = 0.7$, these important NO formation channels reach their maximum contribution. As X_{NH_3} further increases, increasingly more NH_i radicals enter the channel of $\text{NH}_2 + \text{NH} \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$, reducing NO formation. More importantly, NO reduction channels of $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$, $\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$ and $\text{NH} + \text{NO} = \text{N}_2 + \text{OH}$ become increasingly important as X_{NH_3} increases. This explains the levelling out and even a decrease of NO peak concentration as X_{NH_3} increased beyond 0.7 , as shown in Fig. 11.

Note, although data shown in this section include flame cases with three X_{O_2} values of $0.21, 0.30$ and 0.35 , the qualitative behavior among these different X_{O_2} were similar. In this regard, we shall not separately discuss these cases for the effects of X_{O_2} . The consistent trend helped to convince us that the experimentally observed X_{NH_3} effect was not just for a particular case.

3.2.2. Effects of burner exit velocity (V) on NO formation

We established flames with different temperatures on McKenna burner by fixing the reactant composition at $X_{\text{NH}_3} = 0.2$ and $\phi = 0.9$ while varying the burner exit velocity V from $5 - 8 \text{ cm/s}$. Three flame series differing in X_{O_2} were investigated (i.e., the V flame series in Table 1).

The measured and computed axial temperature profiles for the co-fired V flame series are given in Fig. 13a (profile data is shown only for the case $X_{\text{O}_2} = 0.21$, additional data is given in the supplements). Peak flame temperature was retrieved from the profiles to be presented as a function of V in Fig. 13b in which the corresponding neat CH_4 flame case is also included. As can be seen, the experimental and numerical simulation agreed well with each other, confirming the increasing trend of temperature with the increase of V in a nearly linear manner for all the cases tested.

Axial profiles and peak values of NO mole fraction in the V flame series are given in Fig. 14a and b, respectively. Corresponding data for the neat CH_4 flames was also included as dotted lines for comparison purposes. From the profiles a general trend can be observed that NO

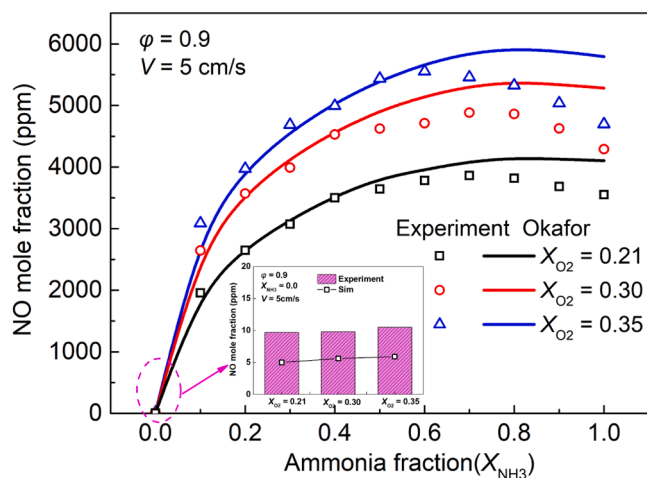


Fig. 11. Variations of peak NO mole fractions in fuel-lean ($\phi = 0.9$) N series flames as a function of X_{NH_3} .

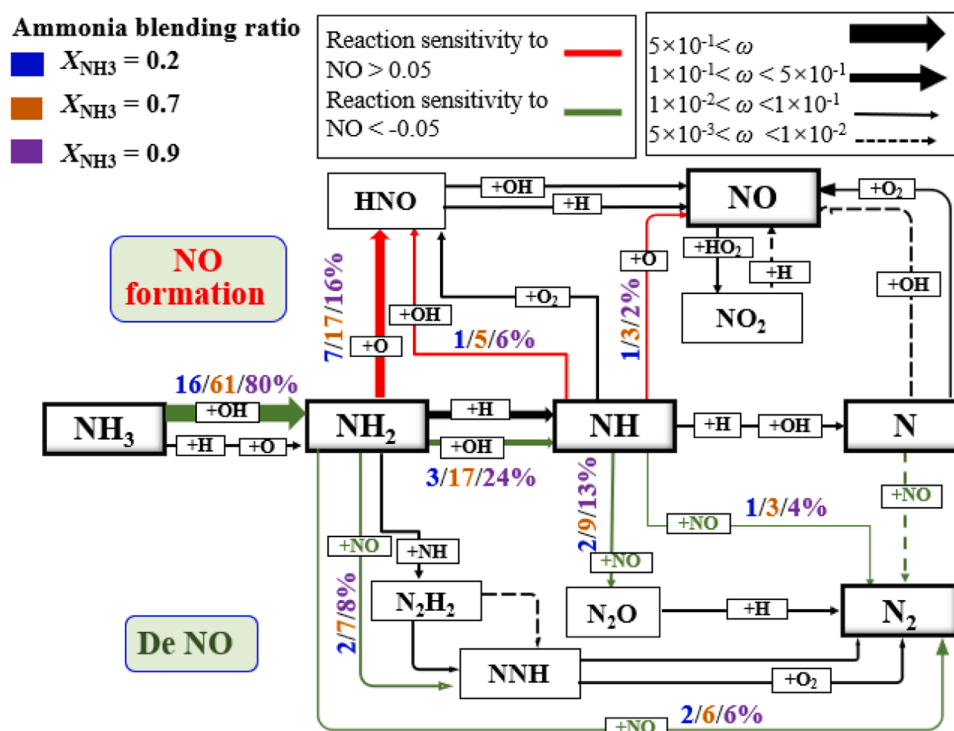


Fig. 12. Reaction pathway analysis for NO formation in the $\text{NH}_3\text{-CH}_4$ cofired flames.

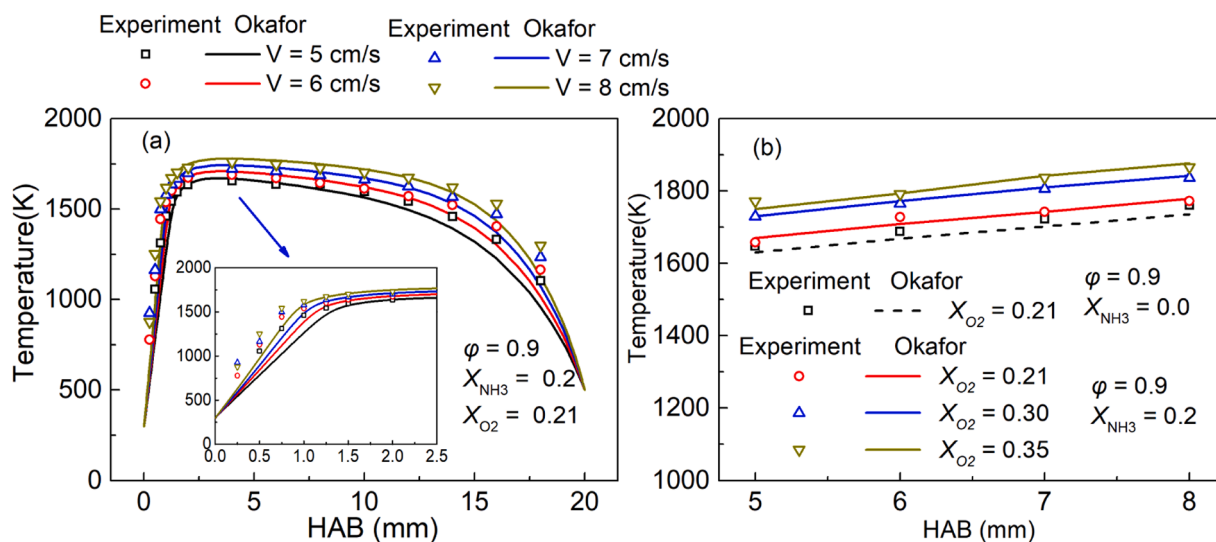


Fig. 13. Measured (symbols) and computed (lines) axial profiles of temperature for the V series flame (a); the peak temperatures are also plotted as a function of V (b).

mole fraction rapid increased before $HAB = \sim 1.2$ mm and then plateaued after peak temperature was reached. Experimental data was seen to agree reasonably well with the numerical simulation.

The data clearly showed that an increase of V , which can be directly translated to an increase of flame temperature, would result in more NO formation. An increase in X_{O_2} similarly enhanced NO formation, mostly due also to the increased temperature (Fig. 13b). Comparing the co-fired and neat CH_4 cases in Fig. 14b, it is noted that the sensitivity of NO formation to temperature is similar, whether for fuel-NO (*i.e.*, co-fired) and prompt-NO dominated (*i.e.*, neat CH_4) cases. This is actually a very interesting result; however, one must be very careful before drawing the conclusion that reducing flame temperature in NH_3 flames would be a NO-inhibiting technique as effective as in conventional

hydrocarbon flames. In fact, the current conclusion is restricted to the relative low temperature range between 1600–1800 K (Fig. 13b). In practical *hydrocarbon* flames where flame temperature can be well above 1800 K, the highly temperature-dependent thermal NO route would be dominant so that many widely recognized industrial NO control techniques such as flue gas recirculation take effect by inhibiting the thermal routes. Whether such techniques can be equally effective in NH_3 flames requires further investigations.

3.2.3. Effects of equivalence ratio (φ) on NO formation

Equivalence ratio is one of the major factors that dictate the combustion and emission characteristics of premixed flames. For NH_3 flames in particular, it has been reported that staged combustion is an effective

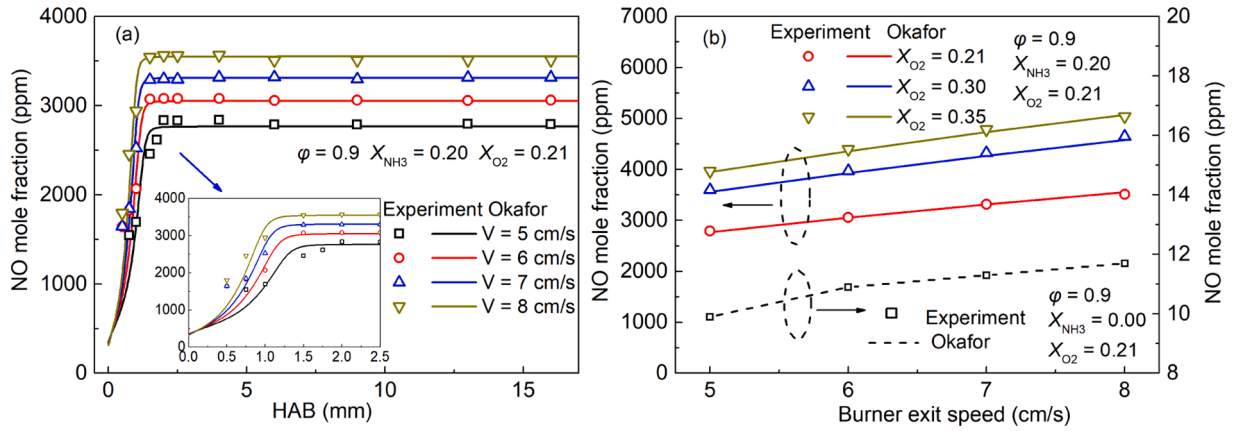


Fig. 14. Measured (symbols) and computed (lines) axial profiles of NO mole fractions for the V series flame (a); the peak NO mole fractions are also plotted as a function of V (b). Data for the neat CH_4 flames are also included as dashed line.

technique for reducing NO emissions in which the most important parameters to optimize are the equivalence ratios in each combustion stages [18,82]. On the other hand, several previous studies have repeatedly shown that existing kinetic mechanisms have notable issues in accurately predicting NO formation in fuel-rich (i.e., $\phi > 1$) NH_3 -containing flames [28,44], indicating that our current understanding on the kinetic evolution from NH_3 to NO may be flawed. In this regard, it is of both theoretical and practical importance to investigate the NO formation characteristics in NH_3 flames over a wide range of equivalence ratios.

In previous sections, we mainly studied fuel-lean flames ($\phi = 0.9$) for which the Okafor mechanism was seen to have satisfactory performances. We shall now focus on kinetic analysis for fuel-rich conditions. Axial profiles and peak values of NO mole fraction are presented in Fig. 15 for the E series flames (data for neat CH_4 flames are also included in Fig. 15b to facilitate a comparison). To make the figure more readable, we show data in the co-fired flame only for cases with $X_{\text{NH}_3} = 0.2$ and $X_{\text{O}_2} = 0.21$; qualitatively similar data for other E series flames listed in Table 1 is provided in the supplements.

The satisfactory prediction of NO with the Okafor mechanism is again seen in Fig. 15a for fuel lean flames with $\phi < 1.0$. However, notable over-prediction is observed for both stoichiometric ($\phi = 1.0$) and fuel-rich conditions ($\phi > 1.0$). In terms of axial profiles, when $\phi \leq 1.0$ both the experiment and numerical prediction show the familiar

trend of an initial rapid increase followed by subsequent plateauing. On the other hand, when $\phi > 1.0$, our experimental profiles show that after NO mole fraction reaching a peak value near the flame front, it exhibits a gradual decrease trend at larger HAB, which indicates the important role of NO consumption reactions in the post-flame region. More specifically, take the $\phi = 1.2$ data in Fig. 15a for example, the NO mole fraction peaked at HAB = 2.0 mm with a value of 466 ppm, and then decreased continuously downstream by 89.6 % to 48 ppm at HAB = 13 mm. Although such a trend is also seen in the numerical results, the quantitative extent of the decrease was underestimated.

The variation of peak NO mole fraction with ϕ can be more clearly seen in Fig. 15b. One striking observation is that peak NO mole fraction increased continuously with ϕ in the neat CH_4 flames, in sharp contrast to the NH_3 - CH_4 cofired flame for which a significant reduction of NO is observed when ϕ increased beyond 1.0. This point shall be further investigated, meanwhile it is also interesting to note the simulation agrees well with the experiments for the neat CH_4 flames under both fuel-lean and fuel-rich cases but notably overpredicts peak NO mole fraction in fuel-rich NH_3 - CH_4 cofired flames. Obviously, this points to our insufficient understanding of the kinetics under NH_3 fuel-rich conditions. The results also suggest that the conventional NO mitigation method of using excessively lean fuel-air mixture for hydrocarbon flames may not be effective for ammonia flames.

Kinetic analysis was performed to provide an explanation to data

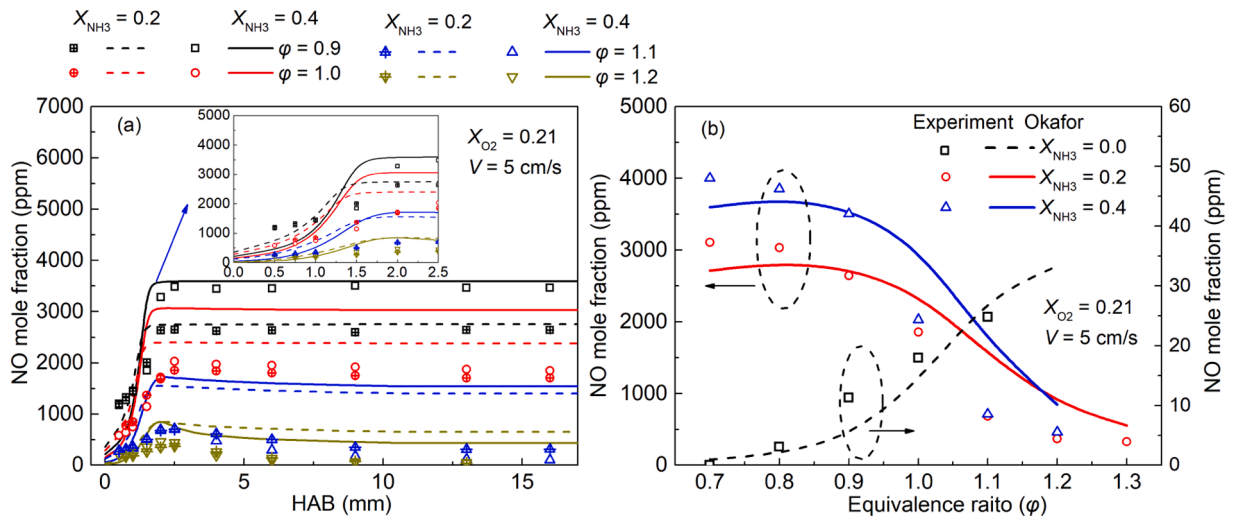


Fig. 15. Measured (symbols) and computed (lines) axial profiles of NO mole fractions for the E series flame (a); the peak NO mole fractions are also plotted as a function of ϕ (b). Data for the neat CH_4 flames are also included as dashed line.

shown in Fig. 15b. ROP analysis of the co-fired flames is shown in Fig. 16 where the main fuel oxidation and NO formation routes can be seen to be significantly affected by φ . In the fuel lean case with $\varphi = 0.8$, NO originates mainly from the route of $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO}$. Additionally, the routes of $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{HNO} \rightarrow \text{NO}$ and $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{NO}$ are also contributing. H abstraction reaction from NH_3 ($\text{NH}_3 \rightarrow \text{NH}_2$) are mainly facilitated by OH/H radicals so that the chain-branching reaction of $\text{O}_2 + \text{H} = \text{H} + \text{OH}$ notably boosts NO formation. As φ increases, the contribution from $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO}$ decreases while more NH_2 radicals would follow $\text{NH}_2 + \text{NH} \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$, avoiding the formation of NO. Additionally, reaction rate for $\text{NH}_i + \text{NO} \rightarrow \text{N}_2$ exhibit a slight decrease but that for $\text{N} + \text{NO} = \text{N}_2 + \text{O}$ notably increase with the increase of φ , indicating the later may play a more important role in NO reduction in fuel-rich flames. In summary, the overall decreasing trend of NO with increasing φ is a result of enhanced NO reduction by various N-containing radicals. Nevertheless, although the Okafor mechanism predicted the correct trend, quantitatively there is a non-negligible discrepancy between the measurement and simulation results (Fig. 15b). Considering the above kinetic analysis, it is speculated the reactions rate for those NO-reducing reactions are underestimated.

A side observation from Fig. 16 is that under fuel-rich conditions ($\varphi = 1.2$), CH_4 oxidation could interact with the oxidation network of NH_3 through $\text{CO}_2 + \text{NH} = \text{HNO} + \text{CO}$, the forward reaction of which contributes to NO formation. An additional interesting point to note is that during the experiments, we found with some surprise that using an oxygen-enriched oxidizer could reduce NO emissions under fuel-rich conditions. We show below the experimental NO mole fraction profiles for fuel-rich co-fired cases ($X_{\text{NH}_3} = 0.2$ in Fig. 17a and $X_{\text{NH}_3} = 0.4$ in Fig. 17b). The comparisons were made between two oxidizer compositions of synthetic air (i.e., $X_{\text{O}_2} = 0.21$) and oxygen enriched air (i.e., $X_{\text{O}_2} = 0.35$). The data showed that the oxygen enriched case had a lower NO emission compared to the synthetic air case. Take Fig. 17a as an example, with $\varphi = 1.1$, although the peak NO value of 864 ppm with X_{O_2}

$= 0.35$ is slightly higher than the value of 707 ppm with $X_{\text{O}_2} = 0.21$, the subsequent NO reduction is significantly faster with the $X_{\text{O}_2} = 0.35$, leading to a significant lower final NO emission of 90 ppm, as compared to the 313 ppm with $X_{\text{O}_2} = 0.21$. Note such a trend is opposite to fuel-lean cases in which higher X_{O_2} would result in a higher NO emission (see Fig. 11). Detailed kinetic investigation and the potential of applying this result as an effective NO control strategy will be pursued in future work, however, it is relevant to point out similar oxygen-enriched NH_3 combustion was pursued to minimize emissions in a very recent work by Liu et al. [83].

3.3. N_2O formation and emission characteristics

In this section, we present a parametric study on N_2O formation in co-fired flames. The results shall also be presented in terms of the effects of X_{NH_3} and φ . Note, it was observed that V has a very weak on measured N_2O profiles so that its effect shall not be further discussed. Instead, a short presentation of the data is given in the supplementary material.

3.3.1. Influence of X_{NH_3} on N_2O distribution

The measured and computed axial profiles of N_2O mole fraction under fuel-lean conditions ($\varphi = 0.9$) are shown in Fig. 18 for X_{NH_3} ranging from 0.2 to 1.0 (Note, N_2O cannot be detected in the $X_{\text{NH}_3} = 0$ case). A general trend is that N_2O rapidly increased and peaked near the flame front before quickly decreasing to almost 0 level in the post-flame zone. The final zero emission of N_2O , irrespective of X_{NH_3} , is consistent with other studies under similar flame conditions [15].

Despite the fact all the X_{NH_3} cases have zero N_2O emissions, the peak values and the spatial distributions of N_2O were dependent on X_{NH_3} . Specifically, with the increase of X_{NH_3} , the peak N_2O mole fraction in the flame continued to increase and the HAB at which such peak value was reached continued to shift downstream. Note, such monotonic increase of peak N_2O mole fraction was different from the trend as observed for

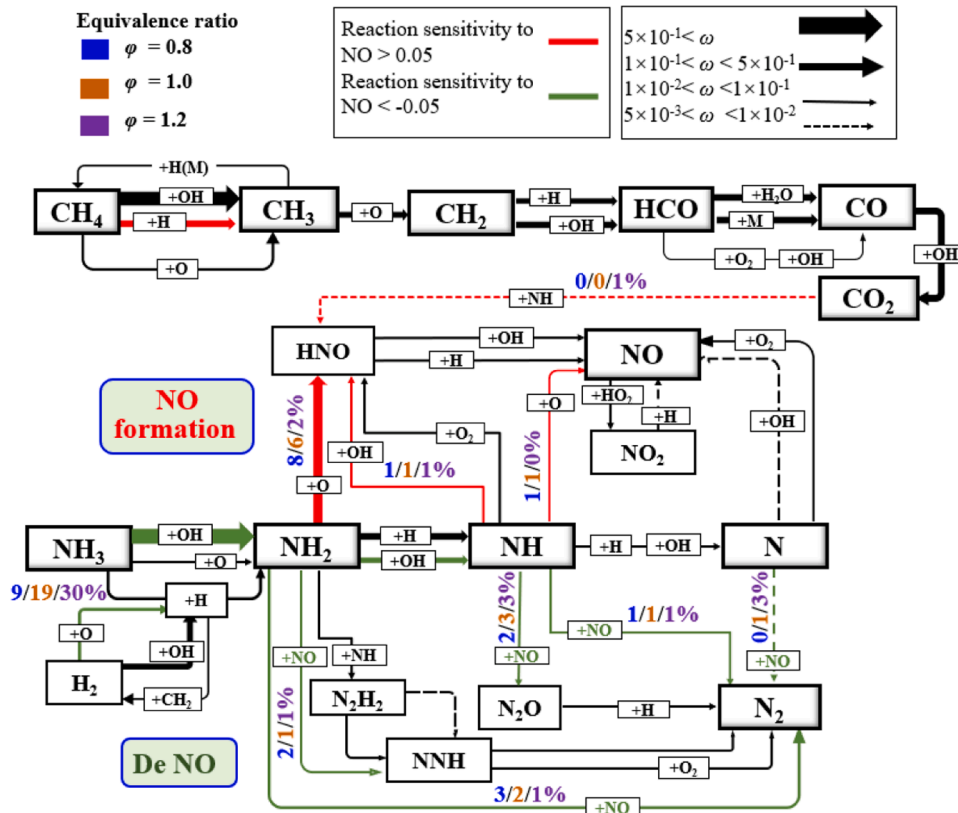


Fig. 16. ROP analysis for NO formation in the E series NH_3 - CH_4 cofired flames with $\varphi = 0.8, 1.0$ and 1.2 .

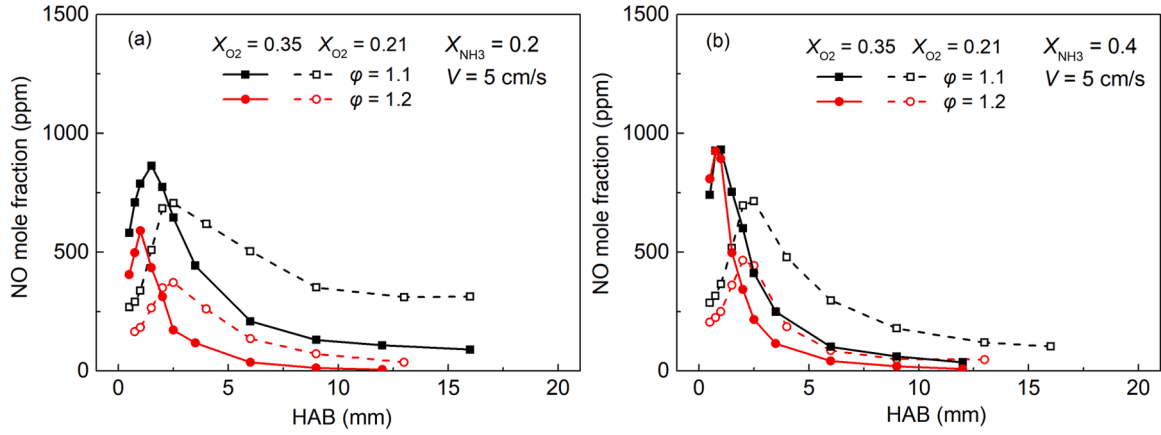


Fig. 17. Effects of oxygen enrichment on axial profiles of NO mole fractions in $\text{NH}_3\text{-CH}_4$ co-fired flames under fuel-rich conditions: $X_{\text{NH}_3} = 0.2$ (a) and $X_{\text{NH}_3} = 0.4$ (b).

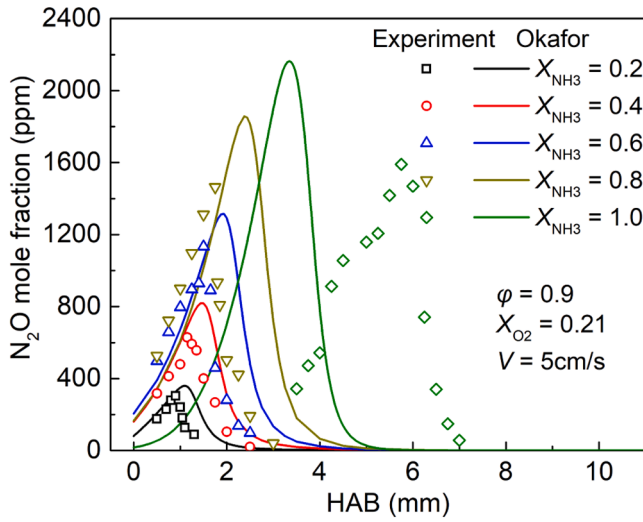


Fig. 18. Effects of X_{NH_3} on axial profiles of N_2O mole fractions in $\text{NH}_3\text{-CH}_4$ co-fired flames under fuel-lean conditions $\phi = 0.9$.

NO mole fractions. The numerical results successfully captured all the qualitative trends although they slightly over-predicted the peak values.

Reaction pathways that involve N_2O in the fuel-lean co-fired flames is given in Fig. 19 for three representative X_{NH_3} values of 0.2, 0.6 and 0.9, where it can be seen the major N_2O formation pathway is $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} + \text{NO} \rightarrow \text{N}_2\text{O}$, with $\text{NH}_i + \text{NO}_2 \rightarrow \text{N}_2\text{O}$ playing a minor role. The contribution of $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$ and $\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ increases with the increase of X_{NH_3} , explaining the increased N_2O peak value. Nevertheless, in the post flame region where radical

concentrations are very much limited all the formed N_2O would be converted back to N_2 , mainly through $\text{N}_2\text{O} + \text{H} = \text{N}_2 + \text{OH}$ (OH and O radicals play a minor role) and direct N_2O decomposition $\text{N}_2\text{O} (+\text{M}) = \text{N}_2 + \text{O} (+\text{M})$ under high temperature conditions.

3.3.2. Influence of ϕ on N_2O distribution

We show in Fig. 20 the effects of ϕ on N_2O formation in co-fired flames with $X_{\text{NH}_3} = 0.2$ across both fuel-lean (a) and fuel-rich conditions (b). The initial upstream and subsequent downstream shift of the peak N_2O location is a result of a corresponding shift of the flame front, which is well captured by the numerical prediction. For fuel-lean conditions, the model also performs well in capturing the peak value N_2O and the small amount left-over of N_2O emission for the $\phi = 0.7$ case. However, for the fuel rich flames, although the model correctly predicted the decreasing trend of peak N_2O value with the increase of ϕ , quantitative overprediction are notable. This seems to be similar with the over-prediction of NO mole fraction under fuel-rich conditions, again demonstrating the necessity for an optimization of the present mechanisms. In addition, we note that when $\phi \geq 1$, the numerical data showed a downstream reemergence of N_2O approaching the end of the computational domain (i.e., near the downstream stagnation plate), which is not observed in the experiment. Unfortunately, at the moment we are not able to identify the origin of this discrepancy.

In discussing the kinetic effects of ϕ , reaction pathways are similar to those given in Fig. 19. Quantitatively, the increase of ϕ increases the contribution of $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$ because of the enhanced radical pool of NH_i radicals. Among the major N_2O consumption reactions, both $\text{N}_2\text{O} + \text{H} = \text{N}_2 + \text{OH}$ and $\text{N}_2\text{O} (+\text{M}) = \text{N}_2 + \text{O} (+\text{M})$ become increasingly important with the increase of ϕ . The overall effect is the increased consumption overwhelms the enhanced formation from $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$, leading to a general decreasing trend of N_2O mole fraction as ϕ increased.

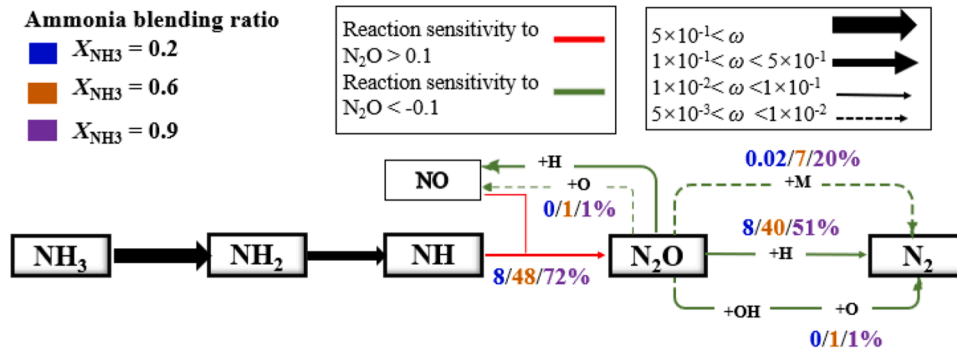


Fig. 19. ROP analysis of N_2O formation in fuel-lean co-fired flames for different X_{NH_3} at 0.2, 0.6 and 0.9.

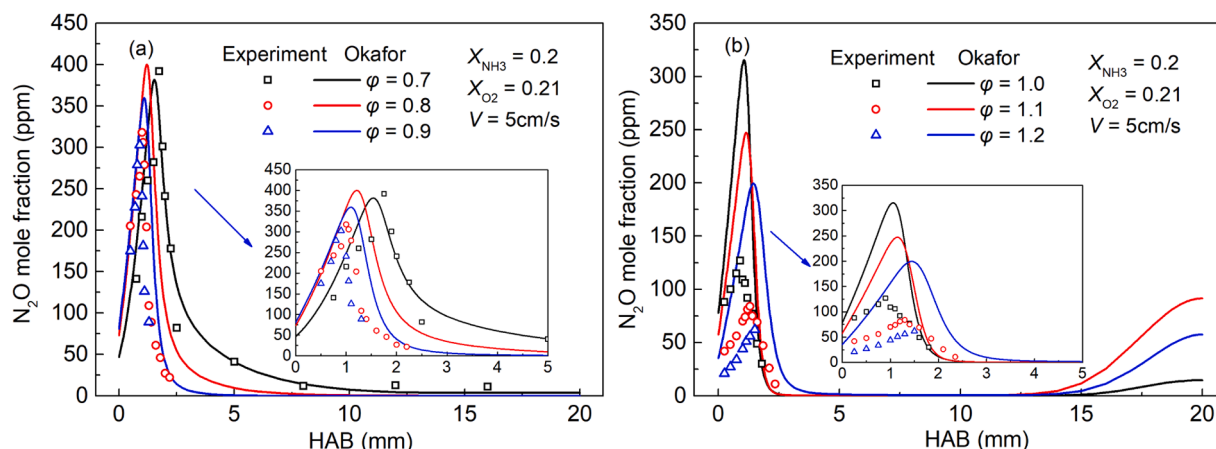


Fig. 20. Effects of ϕ on axial profiles of N_2O mole fractions in $\text{NH}_3\text{-CH}_4$ co-fired flames under fuel-lean $0.7 \leq \phi < 1.0$ (a) and fuel-rich conditions $1.0 \leq \phi \leq 1.2$ (b).

3.4. Concluding remarks

We report $\text{NO}/\text{N}_2\text{O}$ measurement combining laser sensing with microprobe sampling, enabling their quantitative and spatially resolved in $\text{NH}_3\text{-CH}_4$ co-fired laminar premixed flames. The study focuses on the effects of NH_3 fuel ratio, equivalence ratio, temperature and oxidizer composition on the $\text{NO}/\text{N}_2\text{O}$ formation characteristics. Predictive performance of five representative $\text{NH}_3\text{-CH}_4$ co-fired Mech were evaluated. Sensitivity and reaction pathway analyses of $\text{NO}/\text{N}_2\text{O}$ formation were conducted, and the significant discrepancies between experimental and simulation NO formation characteristics under fuel-rich conditions were identified, offering directions for future mechanism optimization. The present work provides extensive spatially-resolved $\text{NO}/\text{N}_2\text{O}$ data in ammonia flames that is urgently needed for the development and validation of NH_3 combustion mechanisms. The result also points to the fact that traditional NO_x mitigation techniques that are popular in combustion of hydrocarbon fuels may not be appropriate in NH_3 flames. The main findings are summarized as follows:

- (1) The addition of ammonia to fuel-lean burner-stabilized CH_4 flames significantly increases NO formation, peaking at $X_{\text{NH}_3}=0.7$ before gradually decreasing. In $\text{NH}_3\text{-CH}_4$ co-fired flames, NO is primarily contributed by fuel-bound nitrogen, with HNO as an important intermediate, and a smaller amount formed via $\text{NH} + \text{O} = \text{NO} + \text{H}$. NO reduction mainly occurs through $\text{NH}_i + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$, with a significantly increased reaction proportion at high X_{NH_3} .
- (2) In the $\text{NH}_3\text{-CH}_4$ co-fired flames, with the increase of the equivalence ratio the peak mole fraction of NO shows a trend of slow decrease until the stoichiometric conditions and then starts to decrease sharply. The increase in equivalence ratio reduces the $\text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO}$ pathway proportion, with more NH_2 radicals undergoing oxidation through $\text{NH}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ and increasing the contribution of reaction $\text{N} + \text{NO} = \text{N}_2 + \text{O}$ under fuel-rich conditions. Notably, experimental data show a significant decrease of NO along HAB when $\phi \geq 1.1$, this trend is enhanced by increased ammonia ratio and oxygen concentration.
- (3) Oxy-fuel combustion enhances $\text{NH}_3\text{-CH}_4$ co-fired flame stability but increases peak NO concentrations at fuel-lean ($\phi < 1.0$). However, it is worth noting that at fuel-rich ($\phi \geq 1.1$) the increase in oxygen concentration can lead to an increase in the rate of NO reduction in the flue gas, effectively reducing NO emissions.
- (4) N_2O rapidly forms and then quickly diminishes within the flame along the axial direction, with peak mole fraction increasing with NH_3 fuel ratio and decreasing with equivalence ratio. Only at low equivalence ratios ($\phi = 0.7$) does the flue gas contain low N_2O

emissions (~ 10 ppm); increasing equivalence ratio result in zero N_2O emissions in the flue gases.

- (5) For NH_3 flames, the persistent high NO emissions under fuel-lean conditions and the interesting NO reduction using oxygen-enriched oxidizers demonstrate the far-reaching consequence of a different NO -formation route from traditional hydrocarbon combustion: certain NO_x mitigation techniques that are popular in combustion of hydrocarbon fuels are not effective, if not counterproductive, in NH_3 flames.

Novelty and significance statement

Despite the wide interests on NO_x control in NH_3 flames, quantitative and spatially resolved $\text{NO}/\text{N}_2\text{O}$ concentration data in canonical laminar ammonia flames are surprisingly scarce. In this work, we demonstrated a novel $\text{NO}/\text{N}_2\text{O}$ measurement method combining microprobe sampling and calibration-free mid-infrared laser absorption spectroscopy and realized spatially-resolved detection with high accuracy and large dynamic ratio. The present study not only provides extensive spatially-resolved $\text{NO}/\text{N}_2\text{O}$ data in ammonia flames that is urgently needed for the development and validation of NH_3 combustion mechanisms, the comparison between neat CH_4 and NH_3 -cofired flames also points to a fact that traditional NO_x mitigation technique that are popular in combustion of hydrocarbon fuels may not be appropriate in NH_3 flames.

CRediT authorship contribution statement

Qing Li: Writing – original draft, Investigation, Formal analysis, Data curation. **Liuhao Ma:** Writing – original draft, Methodology, Data curation. **Jiwei Zhou:** Investigation, Formal analysis. **Jintao Li:** Investigation. **Fuwu Yan:** Resources, Investigation. **Jianguo Du:** Investigation, Data curation. **Yu Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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